

Quiz 3: Qualitative Analysis II — The Chemistry

Review of Session 3. ~15 minutes.

Section A: Formulas

Give the chemical formula:

1. Baking soda
2. Sodium carbonate (washing soda)
3. Calcium carbonate (chalk)
4. Sucrose (table sugar)
5. Starch (cornstarch)
6. Table salt
7. Gypsum
8. Epsom salt
9. Citric acid
10. Vitamin C (ascorbic acid)
11. Sand
12. Sodium acetate

Section B: The iodine test

1. In one or two sentences, explain why starch turns blue-black with iodine.
2. Is the iodine–starch test a chemical reaction? What is the correct term?
3. Amylose turns iodine blue-black. What color does amylopectin (or glycogen) give, and why?
4. What happens to the blue-black color when you heat the mixture? When it cools?
5. Baking soda and baking powder both fizz. One test tells them apart — which, and what result?
6. Vitamin C does something unusual with iodine. What, and why?

Section C: Carbonates and sugars

1. What gas is released when a carbonate reacts with an acid?
 2. Write the word equation for baking soda + vinegar.
 3. Why does chalk fizz more slowly than baking soda?
 4. What does it mean if Benedict's solution turns brick-red?
 5. Does table sugar (sucrose) give a positive Benedict's test? Why or why not?
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Answer Key

Section A

1. NaHCO_3
2. Na_2CO_3
3. CaCO_3
4. $\text{C}_{12}\text{H}_{22}\text{O}_{11}$
5. $(\text{C}_6\text{H}_{10}\text{O}_5)_n$
6. NaCl
7. $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$
8. $\text{MgSO}_4 \cdot 7\text{H}_2\text{O}$
9. $\text{C}_6\text{H}_8\text{O}_7$
10. $\text{C}_6\text{H}_8\text{O}_6$
11. SiO_2
12. $\text{NaC}_2\text{H}_3\text{O}_2$

Section B

1. Amylose coils into a helix; iodine (as triiodide ions) slides inside; the trapped iodine forms a charge-transfer complex that absorbs visible light, so it looks blue-black.
2. No — it's an inclusion complex (a clathrate); the iodine is physically caged in the helix, no bonds change.
3. Red-brown — their branches are too short to hold a long iodine chain.
4. The color disappears (the helix uncoils and the iodine leaves); it comes back on cooling (the helix re-forms).
5. Iodine — baking powder contains cornstarch so it turns blue-black; baking soda alone does not.
6. Iodine goes colorless — vitamin C is a reducing agent and converts the iodine to colorless iodide (this one *is* a real reaction).

Section C

1. Carbon dioxide (CO_2).
2. Sodium bicarbonate + acetic acid $\rightarrow$ sodium acetate + water + carbon dioxide.
3. Chalk is insoluble, so the acid only reaches the surface of each grain.
4. A reducing sugar is present.
5. No — sucrose is a non-reducing sugar (it has no free aldehyde/ketone group) unless it is first hydrolyzed.